UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	INTERACTIVE MULTIMEDIA
COURSE CODE	:	CSC413
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC413 mock paper (NOV 2026).
Answers show the expected marking points; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 15 x 1 mark; Part B = 15 x 1 mark; Part C = 30 x 1 mark; Part D = 4 x 10 marks; total = 100.

4. Questions: 64. Slots: 72. Placeholders: 0. No LLM call.
Questions and answers are for local study use only.

5. Structure derived from 15 unique CSC413 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A - ANSWER KEY

Q	Ans	Statement	Explanation
A1	TRUE	A URL identifies the address of a web resource.	A URL is a resource locator.
A2	FALSE	Interactivity level is irrelevant when choosing an authoring tool.	The required interactivity affects tool choice.
A3	FALSE	A hyperlink can link only to another web page, not to a file.	Links can target files and other resources.
A4	TRUE	Card-based authoring tools are best for messages with a clear beginning and end.	Cards support sequential navigation.
A5	TRUE	HTML is used to structure content on web pages.	HTML marks up web content.
A6	TRUE	Kerning adjusts the space between individual character pairs.	Kerning changes spacing for a specific pair.
A7	TRUE	A prototype is an early model of a product.	Prototypes test ideas before production.
A8	FALSE	Streaming requires the whole file to download before playback.	Streaming can play while downloading.
A9	FALSE	Interlacing displays every line of a video frame at the same time.	Interlacing draws alternate lines.
A10	TRUE	Virtual reality replaces the real environment with a simulated one.	VR creates a simulated environment.
A11	TRUE	A bitmap image is made up of pixels.	Bitmap images use a pixel grid.
A12	FALSE	RAM is a type of permanent storage.	RAM is temporary working memory.
A13	TRUE	Interactivity gives users control over content and flow.	Interactivity is a defining feature of many multimedia products.
A14	FALSE	A multimedia project always uses every media element.	A project selects the elements it needs.
A15	FALSE	A linear presentation allows users to choose their own path.	A linear presentation follows a fixed sequence.

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PART B - ANSWER KEY

Q	Ans	Term	Definition
B1	H	Storyboard	visual sequence plan
B2	G	Keyframe	important frame in an animation
B3	I	Serif	font with small strokes at character ends
B4	C	Multimedia	integration of text, graphics, audio, video and animation
B5	D	Sans serif	font without small strokes
B6	L	Frame rate	frames displayed per second
B7	J	Scanner	input device that digitises documents
B8	B	Plotter	output device for large drawings
B9	N	Card-based tool	authoring tool organised like pages or cards
B10	F	Graphics tablet	input device for drawing
B11	K	Linear multimedia	presentation without user control of sequence
B12	E	Plug and play	automatic device recognition
B13	M	Tweening	generating in-between frames
B14	O	Sampling	converting sound into digital values
B15	A	Chroma key	replacing a colour background

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PART C - ANSWER KEY

Q	Ans	Correct answer
C1	B	Vector
C2	B	CMYK
C3	C	Merging of media and computing
C4	D	Printed output
C5	B	Analog sound to digital values
C6	C	It combines visuals and sound
C7	C	WAV
C8	C	Text, graphics, audio, video and animation
C9	A	A visual sequence plan
C10	C	In-between frames
C11	B	Musical notes and instructions
C12	C	Pixels
C13	B	Permission to use a work
C14	B	Important frames in an animation
C15	D	PNG
C16	A	The user can control content and flow
C17	B	Multimedia with hyperlinks
C18	D	Smooth jagged edges
C19	A	Drawing input
C20	B	Loudness
C21	A	Intended users
C22	D	A video compression family
C23	A	Structure web content
C24	C	Screen display
C25	C	Modelling and rendering
C26	C	One shape into another
C27	A	Drawing each frame by hand
C28	B	Display web pages
C29	D	Card-based
C30	B	Frequency

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PART D - WORKED ANSWERS

QUESTION 1

Topic: authoring interactivity

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

Interactivity support and export formats are important features.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

A medium level with menus and buttons lets users explore without overwhelming them.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

Card-based tools suit structured screens; timeline-based tools suit animation and media sequencing.

(4 marks)

QUESTION 2

Topic: animation

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

GIF and MP4 are suitable; SWF is also acceptable.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

Draw each frame, capture the frames in sequence, then play them back; it gives direct artistic control.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

2-D uses flat drawings such as a cartoon; 2.5-D adds depth to flat artwork, such as parallax layers in a game.

(4 marks)

QUESTION 3

Topic: design project

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

A project manager and an interface designer are important roles.

(2 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

A simple sketch with a title, logo and large topic buttons is sufficient at low fidelity.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

Portability and constant connectivity make mobile devices convenient for games and shopping.

(5 marks)

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QUESTION 4

Topic: graphics

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

Bitmap images use pixels and can pixelate; vector images use mathematical objects and scale cleanly.

(3 marks)

b) Describe TWO factors that affect image quality.

Resolution and colour depth both affect how much detail an image can show.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

Screens use RGB and print uses CMYK; print normally needs a higher resolution for sharp output.

(4 marks)

END OF MARKING SCHEME

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